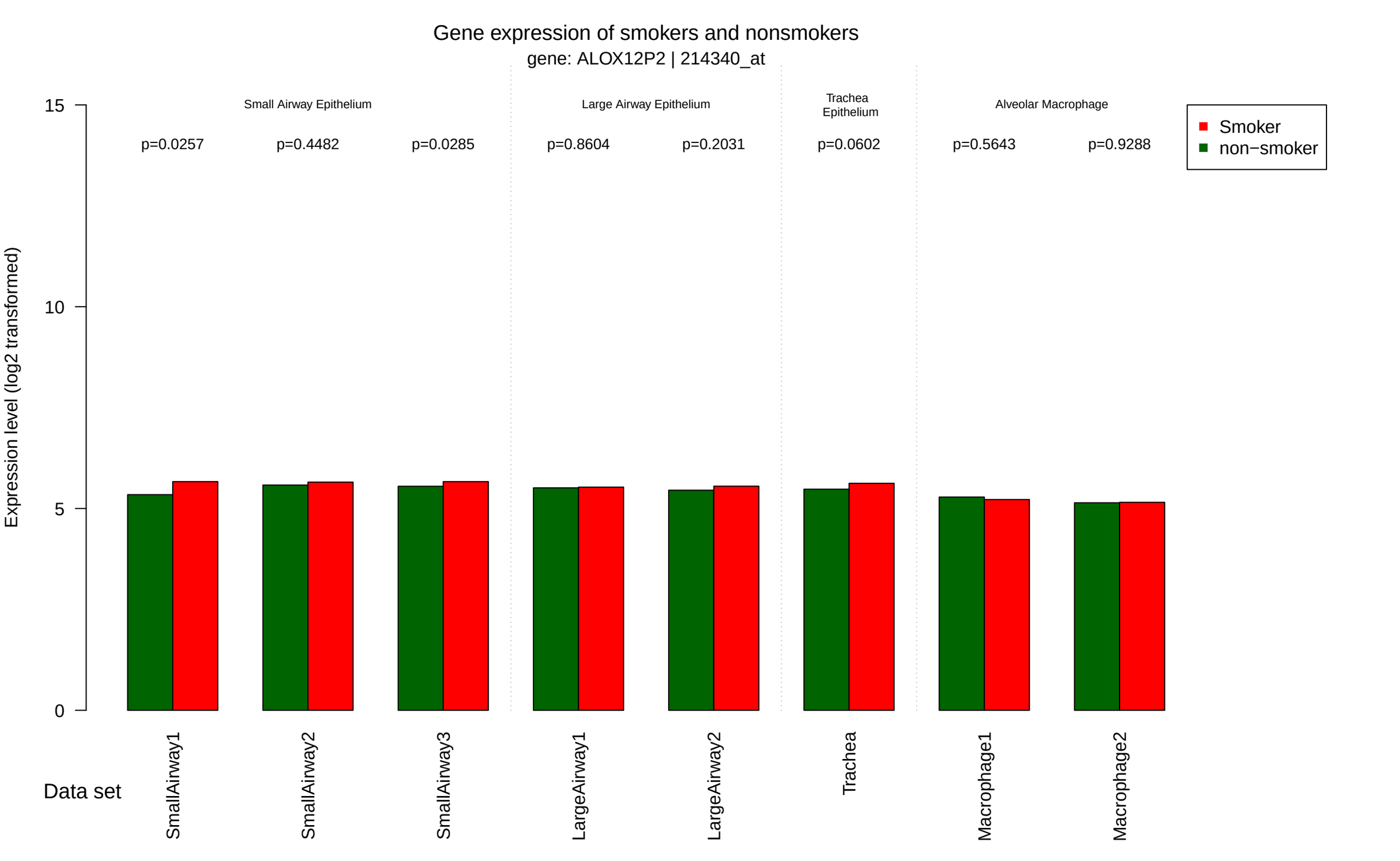
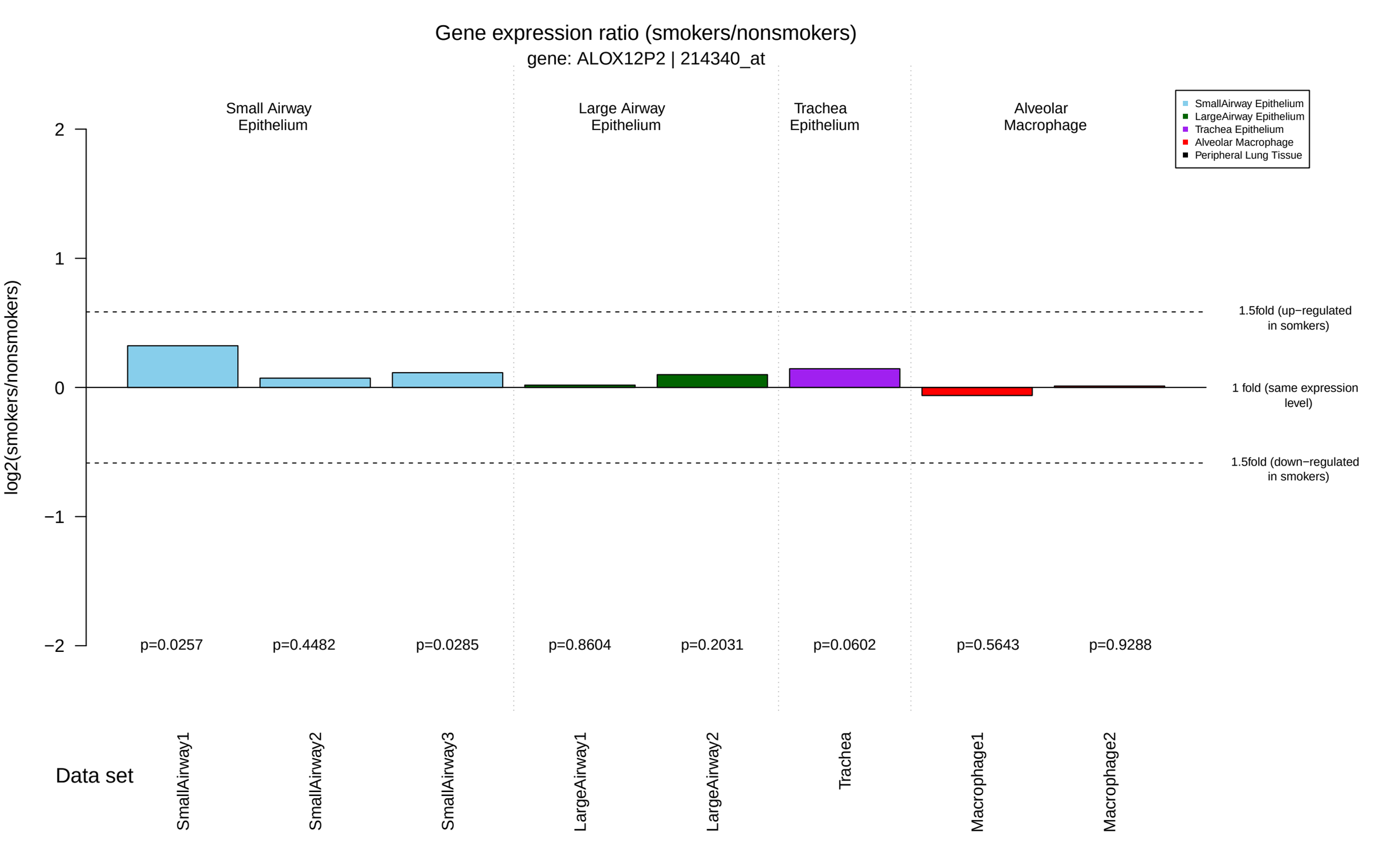
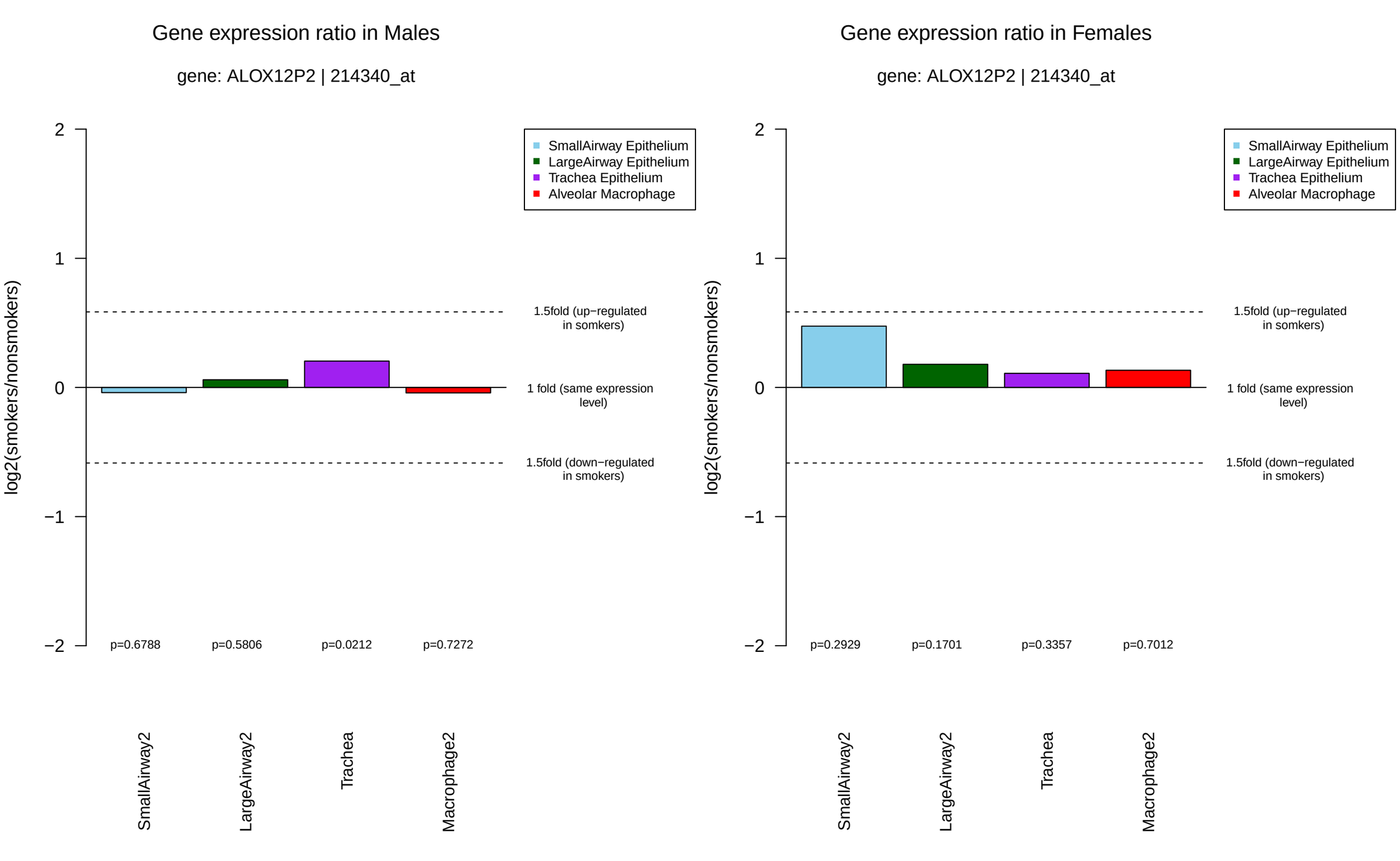
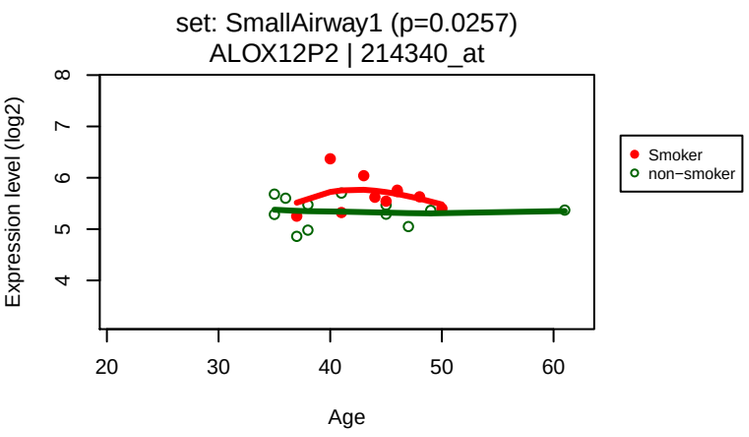
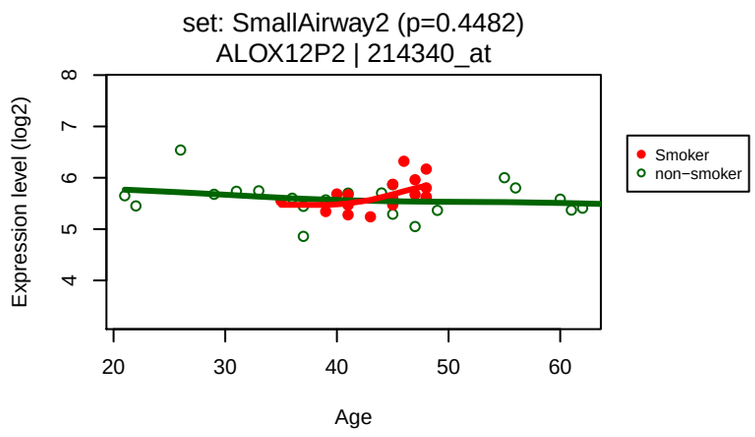
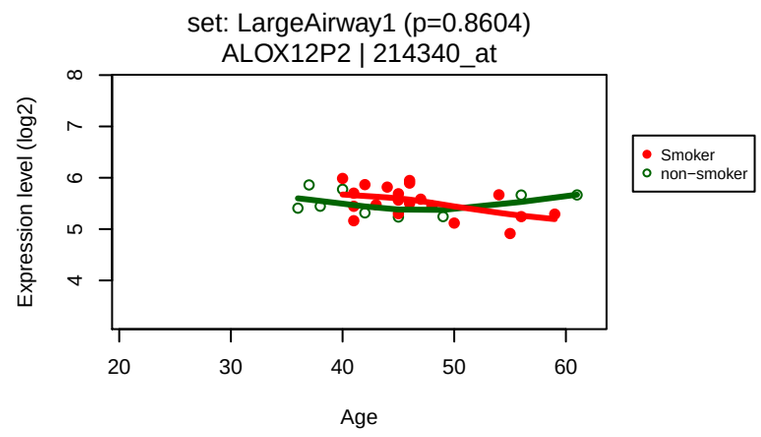

Gene expression of smokers and nonsmokers
gene: ALOX12P2 | 214340_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0257
p=0.4482
p=0.0285
p=0.8604
p=0.2031
p=0.0602
p=0.5643
p=0.9288
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

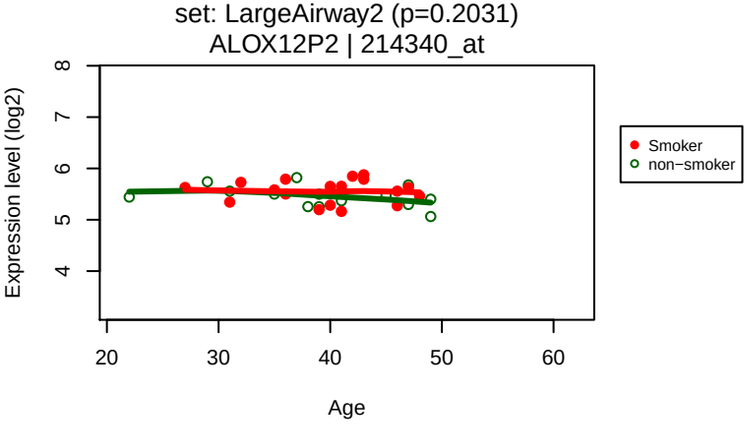

Gene expression ratio (smokers/nonsmokers)
gene: ALOX12P2 | 214340_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0257
p=0.4482
p=0.0285
p=0.8604
p=0.2031
p=0.0602
p=0.5643
p=0.9288
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

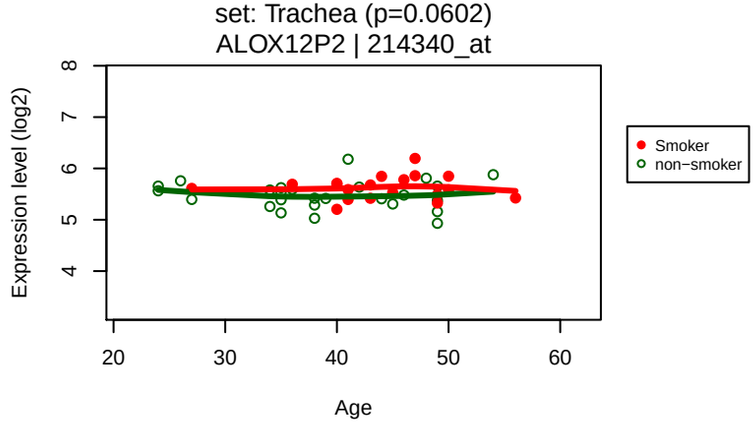

Gene expression ratio in Males
gene: ALOX12P2 | 214340_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6788
p=0.5806
p=0.0212
p=0.7272
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALOX12P2 | 214340_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2929
p=0.1701
p=0.3357
p=0.7012
SmallAirway2
LargeAirway2
Trachea
Macrophage2

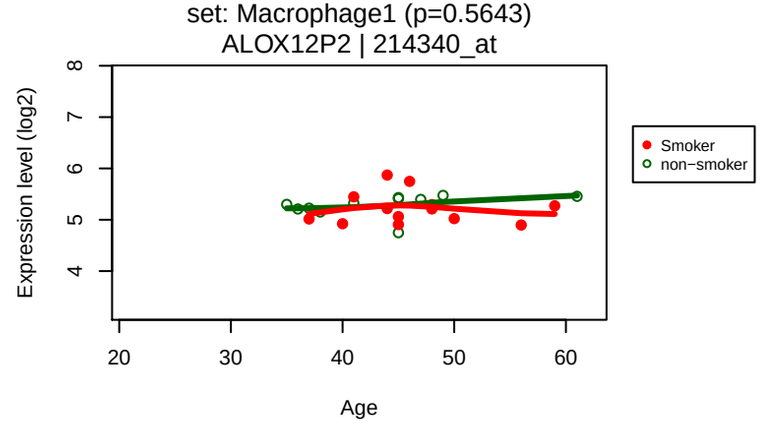

set: SmallAirway1 (p=0.0257)
ALOX12P2 | 214340_at
Expression level (log2)
Age
Smoker
non−smoker

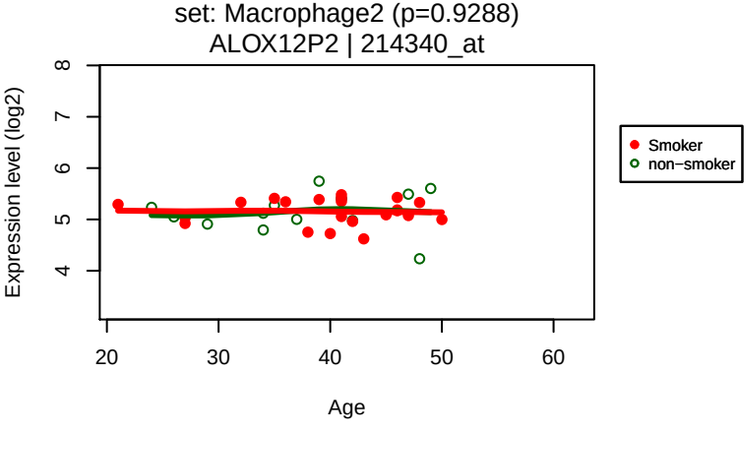

set: SmallAirway2 (p=0.4482)
ALOX12P2 | 214340_at
Expression level (log2)
Age
Smoker
non−smoker

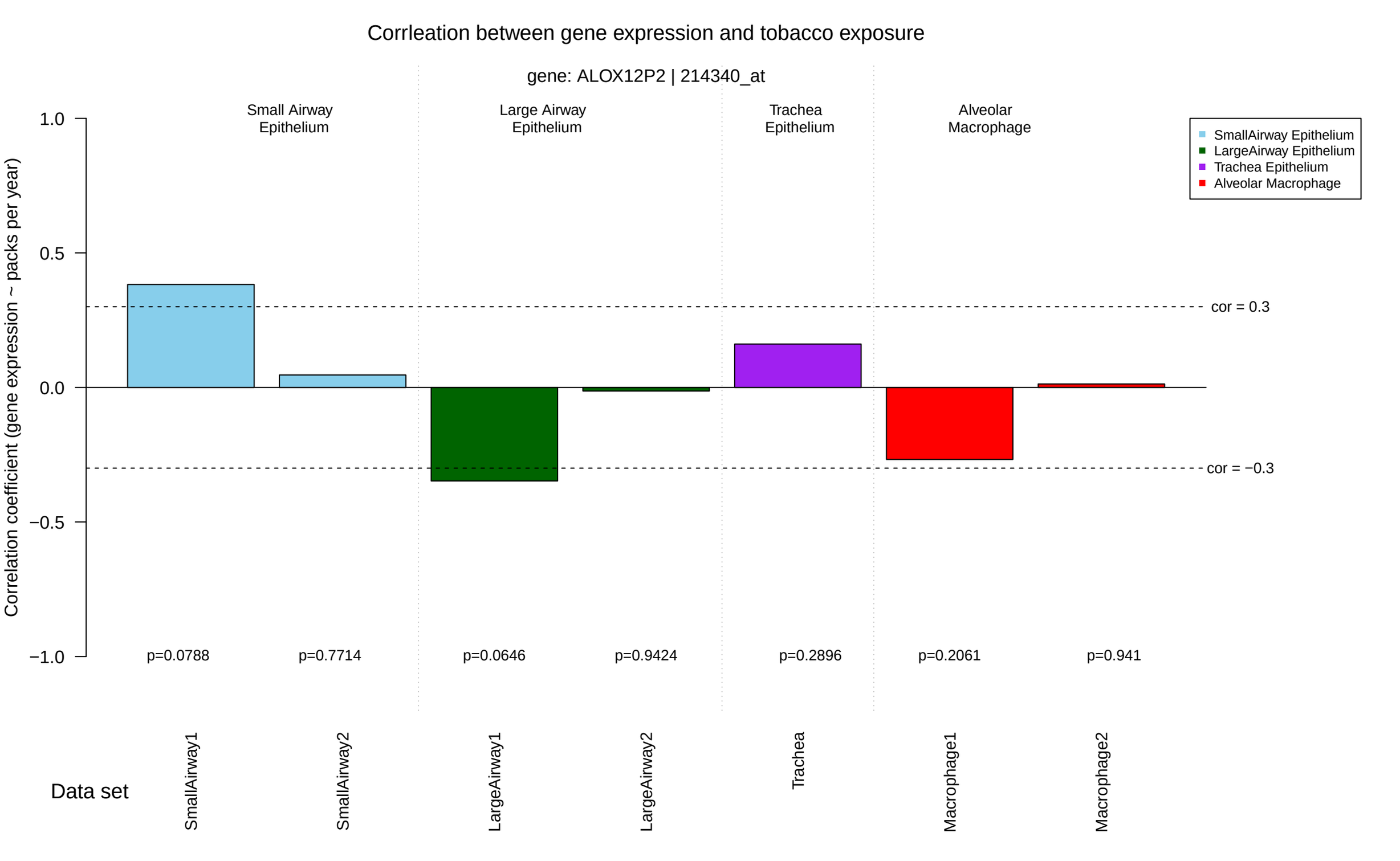

set: LargeAirway1 (p=0.8604)
ALOX12P2 | 214340_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2031)
ALOX12P2 | 214340_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0602)
ALOX12P2 | 214340_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5643)
ALOX12P2 | 214340_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9288)
ALOX12P2 | 214340_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALOX12P2 | 214340_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0788
p=0.7714
p=0.0646
p=0.9424
p=0.2896
p=0.2061
p=0.941
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2